Enhancing Transparency in 3D Reconstruction: An Integrated Metadata and Paradata Framework for Multimodal UAV Documentation

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Abstract

Three-dimensional data has become increasingly important for documenting and sharing cultural heritage, yet significant challenges remain in making 3D models and their associated acquisition processes reproducible, interpretable, and reusable across different contexts. Most existing studies focus mainly on visual quality or technical performance, while the systematic documentation of reconstruction workflows and the semantic enrichment of resulting models remain underexplored. Heritage itself is understood not merely as a physical artifact but as a socially constructed process shaped by communities and institutions (Smith, 2006), making transparent and interpretable documentation essential for long-term scholarly and public engagement. This study presents a structured, metadata-driven workflow for UAV-based 3D documentation of Christian cultural heritage, developed and applied to four churches in the Diocese of Münster, Germany, three historic and one modern structure, selected to represent a range of architectural complexity and material properties. The study addresses two research questions: (1) how can UAV-based 3D reconstruction workflows be systematically documented to ensure reproducibility and transparency, and (2) how can semantic enrichment extend 3D models beyond geometric representation toward Linked Open Data (LOD) compatibility.

The workflow examines three main components. For data acquisition, UAV flight missions are planned and documented using structured metadata and paradata, capturing both technical parameters and the decision rationale behind each mission. This approach draws on recent work demonstrating that flight parameters, including altitude, overlap, and gimbal angle, significantly affect Structure-from-Motion reconstruction quality (Jo et al., 2025), and extends it by formalizing the documentation of those decisions. Second, this study investigates a multimodal reconstruction strategy combining photogrammetric

imagery with LiDAR and polarised imagery to better handle complex architectural features and challenging material properties, including semi-translucent stained-glass windows, where optical interference undermines conventional reconstruction. Third, building on the reconstructed models, this study explores how material properties and structural elements can be linked to the reconstructed geometry through a framework anchored in the CIDOC Conceptual Reference Model (Doerr, 2003), drawing on recent approaches to knowledge graph representation of digitisation paradata (Barzaghi, Moretti, Heibi, & Peroni, 2025) and frameworks integrating RDF metadata with interactive 3D heritage visualisation (Barzaghi, Colitti, Moretti, & Renda, 2025). Figure 1 illustrates UAV image acquisition over a representative church in the Diocese of Münster, and Figure 2 shows the resulting 3D reconstruction using Gaussian Splatting (Kerbl et al., 2023).

Preliminary results across the four church datasets demonstrate the feasibility of the metadata and paradata capture framework under diverse site conditions, with CIDOC-CRM mapping currently being established. The workflow is implemented using Agisoft Metashape for photogrammetric reconstruction, with LOD compatibility assessed through CIDOC-CRM mapping coverage across structural and material elements. Overall, this study highlights the importance of moving toward more integrated and transparent documentation workflows aligned with FAIR principles (Wilkinson et al., 2016), thereby making 3D cultural heritage data more meaningful, interpretable, and easier to reuse across different research contexts.

Fig. 1 UAV image acquisition for Christian cultural heritage documentation.

Fig. 2 3D reconstruction of the church using Gaussian Splatting.

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